

CHARLIE BUSHROW

DATA SCIENTIST | NLP & GENERATIVE AI

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Results-driven Data Science Leader with 8+ years of experience specializing in Generative AI, NLP, and advanced machine learning. Proven track record of designing scalable multi-agent workflows, fine-tuning LLMs for domain-specific applications, and leading cross-functional teams to translate AI research into high-impact business outcomes. Adept at bridging rigorous predictive modeling and scalable cloud architecture to build futuristic enterprise solutions.

TECHNICAL CORE

MACHINE LEARNING & AI

Generative AI, LLM Fine-Tuning, Multi-Agent Workflows, NLP, Text Summarization, NLI, NER, PyTorch, TensorFlow, HuggingFace, Causal Inference (2SLS), Computer Vision.

ENGINEERING & CLOUD ARCHITECTURE

Python, PySpark, SQL, Java, AWS (Bedrock, SageMaker, S3, Lambda, Step Functions), LangChain, LangGraph, VectorDBs, Distributed Systems, Pipeline Automation.

PROFESSIONAL EXPERIENCE

AMAZON Feb 2022 - Present

Data Scientist

Business Intelligence Agent: GenAI SQL Assistant

AWS Bedrock, LLMs, VectorDBs, LangGraph, Python, SQL

- **Architected multi-agent workflows** utilizing LLMs, VectorDBs, and knowledge graphs to accurately interpret domain-specific business context via MCPs and tooling.
- **Spearheaded the research and development** of a generative AI bot that transforms natural language queries into bespoke, executable SQL queries for non-technical stakeholders.
- **Drove cross-functional adoption**, reducing project manager dependencies on data engineering teams by 20% and saving 120 FTE-hours per week, while reducing data request SLAs to minutes instead of hours.

MyVoice: Enterprise NLP & Sentiment Pipeline

AWS (S3, EventBridge, Step Functions, Lambda, Glue, SageMaker, Bedrock), Claude

- **Directed scientific and architectural strategy** for an end-to-end NLP pipeline processing 100K+ unstructured comments monthly as the sole supporting Data Scientist/engineer.
- **Deployed scalable GenAI summarization features** using Claude on Bedrock, implementing advanced prompt engineering and MapReduce techniques to overcome token limits.
- **Fine-tuned domain-specific NLP models** for sentiment detection, multi-class categorization, and PII redaction, transitioning the business to state-of-the-art ML.
- **Partnered with Product and Operations leadership**, translating AI outputs into actionable insights that identified drivers of employee attrition and job satisfaction.

Causal Productivity Modeling & Process Automation

Python, 2SLS, Causal Inference, PyTorch, HuggingFace, NLI, NER

- **Developed a causal inference model** utilizing two-stage least squares (2SLS) regression to quantify the impact of operational changes on employee productivity month-over-month.
- **Engineered scalable pipelines** to transition complex causal research code into resilient production systems with automated data refreshes and performance monitoring.
- **Trained and deployed custom NLP models** for text summarization, natural language inference (NLI), and named entity recognition (NER) to hyper-automate internal workflows.
- **Productionized a regression model** for requisition time-to-fill, implementing dynamic interview ratios that decreased median time-to-fill by 23% and improved hiring manager CSAT by 10%.

BOOZ ALLEN HAMILTON**Jul 2018 - Feb 2022****Lead Data Scientist / Associate****Internal AI Capability & Product Development**

Python, NLP, Dash by Plotly, Predictive Analytics

- **Developed an NLP-powered resource allocation tool** to intelligently match unstaffed team members to open firm-wide requisitions, optimizing internal staffing efficiency.
- **Architected predictive dashboards** using Dash/Plotly to visualize historical hiring trends and forecast time-to-hire metrics across specific client and clearance categories.
- **Reduced unbilled bench time and accelerated resource deployment** by empowering leadership with dynamic, data-driven visibility into talent availability and internal talent pipelines.

Air Force Readiness Analysis and Decision Support (AFRADS)

Predictive Modeling, Sensitivity Analysis, Big Data Aggregation

- **Single-handedly informed a \$500 million reallocation** of US Air Force defense funding by developing advanced predictive models and conducting high-stakes sensitivity analyses.
- **Translated complex, large-scale readiness data** spanning the Flying Hours Program, Weapons System Sustainment, and OpsPersTempo into clear, decisive go/no-go strategies for senior military leadership.

Intelligence Community Data Science Instruction & Mentorship

Technical Leadership, Machine Learning, Knowledge Sharing

- **Spearheaded upskilling initiatives** for the Intelligence Community, acting as a Subject Matter Expert in the design and facilitation of comprehensive data science and machine learning curricula.
- **Retained the top instructor rating** across the contract by translating overall technology goals into personalized course material, mentoring technical teams to achieve innovation excellence.

Aerospace Computer Vision & Simulation Modeling

Python, fast.ai, React.js, Java, AnyLogic, Monte Carlo Simulation

- **Built a high-accuracy computer vision application** (>98% accuracy) for the classification of 20+ aircraft models. Handled the full lifecycle from developing fast.ai models to integrating them into a React.js web application.
- **Developed complex agent-based simulations** using Python, Java, and AnyLogic to model flight line operations, empowering senior leadership to run data-driven Monte Carlo "what-if" scenarios for facility and scheduling optimization.

EDUCATION

Georgia Institute of Technology | Master of Science in Analytics

In Progress (4.00 GPA)

Virginia Tech | Bachelor of Science in Industrial & Systems Engineering

CERTIFICATIONS**AWS Certified:** Machine Learning Specialty | Solutions Architect Associate | Cloud Practitioner**Additional:** Microsoft Azure Fundamentals | ICAgile Certified Professional